Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

Andrew H. Bond
San José State University
andrew.bond@sjsu.edu ORCID 0009-0003-2599-6158

Working draft — v0.6

Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [14]. We show the geometry does not vanish; it migrates entirely into the *angular* coordinate. Writing each node's low-mode embedding in polar form (magnitude × direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is the row-normalization of Ng–Jordan–Weiss spectral clustering [8] applied to the commute-weighted ($1/\sqrt{\lambda_k}$) embedding (the weighting matters: §5.6), whose existing theory concerns cluster separation [12]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension $d \geq 2$, and **conjecture**, with strong empirical support, that the angular coordinate remains rank-faithful to geodesic distance—flat in both mode-count and graph size where the raw embedding decays. The basis is **substrate-independent**: it preserves geodesics on any graph carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and degrades on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx$ 1–3.6), fidelity falls approximately linearly with intrinsic dimension. Refined toward the continuum (N up to 5×10^4) at a fixed mode budget, the basis *sharpens* on sampled Riemannian substrates (angular fidelity monotone increasing across the five sizes) yet *decays* below a pre-registered floor on a coordinate-free Wolfram-model manifold—a budget-relative dissociation (it inverts when the budget is scaled) that recasts the angular coordinate as a *diagnostic* for whether a substrate's fine structure stays manifold-like under refinement.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [5]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [14]. Yet the most successful spectral-clustering algorithm [8] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [12, 13]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph's geodesic structure, and why.**

Our answer is a clean decomposition. Write each node's low-mode embedding in polar form, magnitude × direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg's theorem kills; the *direction* carries the graph's geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This "keep the scale-invariant direction" operation recurs beyond clustering—for example in polar-coordinate KV-cache quantization, which transfers the scale-invariant direction and quantizes the radius separately [10]. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [17], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated—and, we show in §5.11, the same angular diagnostic reads out on a real semantic embedding space, a language model's moral-text vectors.

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *corollary* (for the full embedding, from [14]) with a spectral-gap bound on the truncation error (Proposition 1) and an empirically-supported Angular-Preservation *conjecture* now with first metric-distortion evidence; (2) a measurement with controls (§5); (3) universality across five substrate types with a two-factor small-world screen (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); (5) a cross-domain probe on a language-model semantic embedding (§5.11); and (6) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [5] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [14] proved the commute distance of a large geometric graph degenerates, $R(i,j) \to 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. They also propose an *amplified* commute distance as a correction, the natural competitor to ours; rather than repair the radius we discard it and keep the angle. **Row-normalization.** Ng–Jordan–Weiss [8] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [12] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k-means' post-normalization success; consistency of spectral clustering is treated in [13]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. The same normalization removes degree as a nuisance in degree-corrected block models—SCORE [23] cancels it through eigenvector ratios, and spherical spectral clustering under the DCBM [24] projects to the sphere—but those are community-recovery guarantees, not geodesic ones. The embedding Ψ itself is Rustamov's Global Point Signature [22] from shape analysis, and the magnitude we discard is used in its own right for cluster and anomaly detection [25]; we keep the direction and give it a metric reading. **Emergent geometry and observer theory.** In the Wolfram model [15, 16, 6] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [17]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [7, 11, 9]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k), $0 = \lambda_0 < \lambda_1 \leq \cdots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = \left(v_1(i)/\sqrt{\lambda_1}, \ldots, v_m(i)/\sqrt{\lambda_m}\right)$. Write $\Psi_i = r_i\,\hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$. [1]

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime $d \geq 2$ refers to this measured intrinsic dimension; §5.4 confirms the $d = 1$ boundary behaves as predicted.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 3$ satisfying the conditions of [14] (the $d = 2$ case is marginal: at the recurrent boundary the resistance carries logarithmic corrections, matching the borderline behaviour we measure in §5.5), and let Φ_i be the* full *commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L^+_{ii}$. As $n \to \infty$ the resistance distance degenerates, $R(i,j) = L^+_{ii} + L^+_{jj} - 2L^+_{ij} \to 1/k_i + 1/k_j$. Summing over j and using the gauge $\sum_j L^+_{ij} = 0$ (from $L\mathbf{1} = 0$) gives $\sum_j R(i,j) = n\,L^+_{ii} + \operatorname{tr} L^+$; matched against $\sum_j(1/k_i + 1/k_j) = n/k_i + \sum_j 1/k_j$ this forces $L^+_{ii} = 1/k_i + C$ with C independent of i, and taking the trace of that relation gives $C = 0$ (the sum is legitimate because the vL–R–H limit is uniform over the n pairs). Hence $L^+_{ii} \to 1/k_i$, so the radius satisfies $\|\Phi_i\| \to 1/\sqrt{k_i}$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting needs (i) the normalized analogue of the degeneracy [14] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 1). Numerically the degeneracy holds tightly: $\rho(r_i, 1/\sqrt{k_i}) = 0.92$–$0.99$ and radius-only geodesic preservation $\rho \leq 0.08$ (§5.5).

Proposition 1 (Truncation is spectral-gap bounded). *Let $\Psi^\infty_i = \left(v_k(i)/\sqrt{\lambda_k}\right)_{k=1}^{n-1}$ be the full symmetric-normalized commute-style embedding and Ψ^m_i its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi^\infty_i\|^2 - \|\Psi^m_i\|^2 \leq \frac{1}{\lambda_{m+1}}, \qquad 0 \leq \|\Psi^\infty_i - \Psi^\infty_j\|^2 - \|\Psi^m_i - \Psi^m_j\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda^{-1}_{m+1}\sum_{k>m} v_k(i)^2$ and $\lambda^{-1}_{m+1}\sum_{k>m}(v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}_{k=0}^{n-1}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for

[1] We call Ψ "commute-style" rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n-1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance $R(i,j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

$i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m}(v_k(i)-v_k(j))^2 \leq \sum_k(v_k(i)-v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2\sum_k v_k(i)v_k(j) = 2$. These identity sums run over all $k = 0, \ldots, n-1$, including the trivial mode $k = 0$ that Ψ omits; it sits only in the subtracted low-mode partial sums, and dropping its nonnegative terms can only tighten the tail bounds. $\square$

Two consequences. First, the error is controlled by the inverse spectral gap $1/\lambda_{m+1}$ alone. This is *not* an n-independent absolute bound: on a fixed-degree neighbourhood graph refined toward the continuum λ_{m+1} itself shrinks, so $1/\lambda_{m+1}$ grows. What it controls is the truncation error *relative to the embedding scale*, which is fixed by the same low eigenvalues; since the angular and rank statistics we report are invariant to global scale, that relative control is what transfers, and is why the continuum refinement of §5.10 does not degrade. Second, the deleted content is high-frequency (large-λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5), so the bound formalizes why aggressive truncation does not lose the geometry—it discards a spectral-gap-bounded, high-λ remainder. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 1 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 1.

Conjecture 1 (Angular Preservation). *For ε-graph or k-NN samples of a compact Riemannian manifold satisfying the density conditions of [3, 21] (an explicit class, not one defined post hoc by our own estimators), row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise distances among the resulting directions remain* rank-faithful *to geodesic distance—Spearman correlation bounded away from zero (empirically ≈ 0.9)—uniformly in the mode-count m and size n. (We reserve "rank-faithful" for Spearman $\geq \rho_0 > 0$ over the stated class; the diagnostic reading—using the ω-screen and dimension spread to decide whether a* given *graph lies in the class—is a separate corollary-in-use, kept out of the conjecture to avoid circularity.) A stronger, metric form of the conjecture—that the angular map is bi-Lipschitz to geodesic distance with constants uniform in m and n—is consistent with our data but untested; our evidence is rank-based and cannot by itself bound metric distortion.*

Metric evidence on the torus. The metric form is no longer entirely untested. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale, the angular map has a robust empirical bi-Lipschitz constant $\kappa \approx 2.6$ (ratio of the 97.5th to 2.5th percentile of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs), tighter than the full commute embedding (3.1), an equal-size random-mode basis (8.3), or the magnitude coordinate (125); and its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.07, versus 0.995 for random modes. This is genuine metric distortion, not rank correlation—one substrate, but a well-understood one. (The full-range constant is inflated by the few near-antipodal pairs the torus folds together, hence the percentile-robust κ.) A uniform-in-(m, n) bi-Lipschitz bound remains the open metric conjecture; this is one datapoint toward it.

3.1 Toward a proof of Conjecture 1

Proposition 1 controls truncation; the remaining question is whether row-normalization preserves the geometry the low modes carry. It does, and the mechanism is an exact identity.

Lemma 1 (Normalization preserves bi-Lipschitz geometry)**.** *For nonzero $u, v \in \mathbb{R}^m$ with $\hat{u} = u/\|u\|$,*

$$\|\hat{u} - \hat{v}\|^2 = \frac{\|u - v\|^2 - \big(\|u\| - \|v\|\big)^2}{\|u\|\,\|v\|}.$$

If $F : (X, d) \to \mathbb{R}^m$ satisfies $L^{-1} d(x, y) \leq \|F(x) - F(y)\| \leq L\, d(x, y)$ with $\rho^- \leq \|F\| \leq \rho^+$ and radius Λ-Lipschitz, then $\hat{F} = F/\|F\|$ obeys $\|\hat{F}(x) - \hat{F}(y)\| \leq (L/\rho^-)\, d(x, y)$, and if $\Lambda < 1/L$,

$$\|\hat{F}(x) - \hat{F}(y)\| \;\geq\; \frac{\sqrt{L^{-2} - \Lambda^2}}{\rho^+}\, d(x, y).$$

Proof. The identity combines $\|\hat{u} - \hat{v}\|^2 = 2 - 2\langle u, v\rangle/(\|u\|\|v\|)$ with $\langle u, v\rangle = \frac{1}{2}(\|u\|^2 + \|v\|^2 - \|u - v\|^2)$. The upper bound uses numerator $\leq \|u - v\|^2 \leq L^2 d^2$ and $\|u\|\|v\| \geq (\rho^-)^2$. For the lower bound, $(\|u\| - \|v\|)^2 \leq \Lambda^2 d^2$ and $\|u - v\|^2 \geq d^2/L^2$ give numerator $\geq d^2(L^{-2} - \Lambda^2)$; divide by $\|u\|\|v\| \leq (\rho^+)^2$. □

The condition $\Lambda < 1/L$ is a quantitative radial degeneracy: the radius must vary *slower* than the embedding moves. As $\Lambda \to 0$ (radius $\to$ constant, the limit of Corollary 1) the lower constant tends to $1/(L\rho^+)$ and normalization becomes a pure scaling. On the 2-torus this holds pointwise: the identity is exact to 10^{-15}, and the radius Lipschitz constant $\Lambda = 0.0918$ sits only marginally under the map's lower stretch $A = 0.0920$. The margin is razor-thin, so Lemma 1's *worst-case* lower constant $\sqrt{L^{-2} - \Lambda^2}/\rho^+$ is near-vacuous here—but that crude bound is not the operative one. Proposition 2 shows the lower stretch is the pointwise *tangential* component of dF, which stays bounded away from zero (0/9200 local pairs are radially degenerate, median Λ five times below A), giving a measured local angular bi-Lipschitz constant of 2.2 that is an empirical fact, not a consequence of the crude lemma. Lemma 1 links the two halves of the principle and reduces the conjecture to standard ingredients.

Proposition 2 (Exact angular distortion)**.** *Let F be differentiable with $r = \|F\| > 0$ and $\hat{F} = F/r$. Taking $y \to x$ in Lemma 1 gives, for every tangent vector v,*

$$\|d\hat{F}(v)\|^2 = \frac{\|dF(v)\|^2 - dr(v)^2}{r^2} = \frac{\|dF(v) - \hat{u}\, dr(v)\|^2}{r^2}, \qquad dr(v) = \langle dF(v), \hat{u}\rangle.$$

The angular stretch in a direction v is *exactly* the tangential part of $dF(v)$—the component orthogonal to the radius—divided by r. So $\hat{F}$ is a local bi-Lipschitz immersion *iff* dF is nowhere purely radial ($dF(v) \not\parallel F$ for $v \neq 0$), the lower stretch being $\inf_v \|dF(v) - \hat{u}\, dr(v)\|/(r\|v\|)$. This sharpens Lemma 1: the crude $\Lambda < 1/L$ merely forces the radial term $dr(v)$ below the total $\|dF(v)\|$, whereas the proposition shows what is really required is that the embedding's velocity never aligns with its position. Lemma 1 and Proposition 2 link the two halves of the principle and reduce the conjecture to standard ingredients.

Theorem 1 (Conditional local form of Conjecture 1)**.** *Let M be a compact d-manifold whose graph Laplacian spectrally converges to M's ([3] for ε-graphs, [21] for k-NN graphs), and fix a mode-count m at a spectral gap of M. Assume **(H1)** the truncated eigenmap $x \mapsto (\varphi_k(x)/\sqrt{\mu_k})_{k=1}^m$ is L_0-bi-Lipschitz onto its image on geodesic balls of radius r_0—the quantitative truncated-eigenmap embedding property of [18, 19, 20]—and **(H2)** the associated truncated radius is Λ_0-Lipschitz with $\Lambda_0 < 1/L_0$ and lies in $[\rho^-, \rho^+]$. Then there exist constants $0 < A \leq B$ independent of n such that, with high probability as $n \to \infty$, the graph angular map obeys $A\, d_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq B\, d_M(x, y)$ for all sampled x, y with $d_M(x, y) \leq r_0$.*

Proof sketch. By (H1) the continuum eigenmap is locally bi-Lipschitz; by (H2) and Lemma 1 its normalization, the continuum angular map, is locally bi-Lipschitz. The transfer to the graph needs *sup-norm* eigenvector convergence (so that normalization survives where $\|F\|$ is smallest and the graph radius inherits $[\rho^-, \rho^+]$)—the rates of [21] rather than generic L^2 consistency—with "high probability" taken over the sampling model of those results; Proposition 1 then bounds the m-truncation. We give this as a sketch: the uniform-in-m control of (H1) and the sup-norm rate bookkeeping are the delicate steps, not fully carried out here. □

What remains is delimited. **(H1)** is subtle: Jones–Maggioni–Schul [4] guarantee bi-Lipschitz charts only from a *well-chosen* d-subset of eigenfunctions per ball, because the lowest modes can degenerate at critical points; the truncated-eigenmap results [18, 19, 20] give quantitative control but the constants' uniformity in m (our empirical "flat in m," §5.5) is open. Low-spectrum multiplicity compounds this—the square torus's first nontrivial eigenvalue has multiplicity four, so a truncation *inside* a multiplet ($m = 5$) is basis-dependent; distances are well-defined only for whole eigenspaces, so m should be taken at a spectral gap, or the map phrased through spectral projectors. **(H2)** is, by Proposition 2, sharper than radius flatness: what is needed is that the eigenmap's velocity $dF(v)$ is nowhere purely radial—a transversality the immersion (H1) makes generic, and the truncated analogue of von Luxburg–Radl–Hein. It holds on the torus (tangential lower stretch bounded, local constant 2.2); whether the eigenmap image ever develops a radial fold on some manifold is the sharp remaining question. Globally a torus admits no bi-Lipschitz Euclidean embedding, so no global metric statement can hold; the global claim is necessarily the rank-faithful one this paper measures. Conjecture 1 is thereby reduced to a local-chart hypothesis (H1) and a radius-flatness hypothesis (H2), neither of which mentions the angle directly.

Status. Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Corollary 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88$–0.90) even on quasi-1D graphs where Corollary 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. That mechanism is analyzable rather than mysterious: if the low L_{sym} eigenvectors converge to Laplace–Beltrami eigenfunctions [2, 3] and eigenfunction coordinates furnish locally bi-Lipschitz charts of the underlying manifold [4], the angular map's geodesic fidelity should follow from the chart geometry—the natural route to a proof of Conjecture 1, and this paper's central open problem. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i.

Dimension gating. Corollary 1 is a $d \geq 2$ phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread ⇒ clean manifold, and the agreed value is our measured intrinsic dimension d. **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius only), *euclid* (classical MDS of the graph geodesics, i.e. Isomap; the table of §5.3 reports it at dim$=$2). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ for the substrate benchmarks, 200 for the torus metric tests (§3.1) and the cross-domain probe of §5.11—11,175 and 19,900 pairs respectively). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers); the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, m=10: 0.914 ± 0.009 over five draws). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v), u the roll angle and v the height—carried by a random orthogonal map into $\mathbb{R}^{12}$ with isotropic noise (σ=0.03) and connected by a k=10 nearest-neighbour graph on n=2000 ambient points. It stands in for sequential/embedding-derived data (each item a point on a low-d semantic manifold, consecutive items nearby), not a specific text corpus.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3. On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag.×dir.)	**angle only**	magnitude only
5	0.761	**0.926**	0.025
10	0.650	**0.913**	0.035
20	0.550	**0.891**	0.028
40	0.517	**0.928**	0.034

The angle carries the geometry (≈ 0.92, flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2); averaged over ten random draws $\rho = 0.08 \pm 0.16$, individual draws occasionally catching a low mode and spiking to ~ 0.4.

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only ≈ 1.5D manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle-$\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

5.3 Universality across substrates

substrate	meas. d	angle-ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0*	0.45	0.00	—

Substrates with genuine low-d Riemannian geometry all show the effect, independent of graph origin; the metric degrades where geometry is Lorentzian ($\rho = 0.57$, a degradation not a failure) or destroyed. *For the small-world graph the three dimension estimators disagree sharply (ball 3.7,

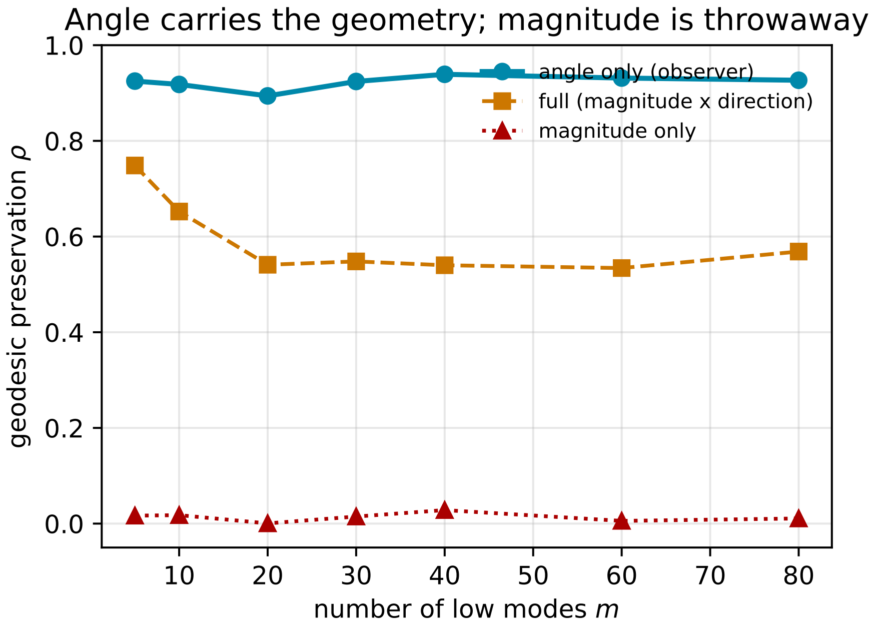

Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) as a function of mode-count m. Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m.

spectral 3.6, effective-rank 4.3; spread 0.77 versus 0.14–0.51 across the clean substrates); by our own manifold-grade criterion this graph is *not* a manifold, and the low angle-ρ is the predicted failure, not an exception. The two-factor screen below makes this quantitative.

Two caveats keep the universality claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in m and n (§5.5). That 0.67 is Isomap forced into $\dim = 2$; matched to $\dim = m$ it reaches ≈ 0.97 and edges the angle even on the torus—but Isomap is *given* the geodesic matrix as input, whereas the angular basis recovers it from the spectrum alone, so the two solve different problems. Universality here means the angular basis never *loses* the geometry on a genuine low-d Riemannian substrate—not that it dominates every baseline on every substrate. Second, the swiss-roll is a manifold-embedding case (a smooth 2D manifold carried into $\mathbb{R}^{12}$): the angular basis is the same whether the manifold comes from hypergraph rewriting or from points sampled on a continuous manifold. No sequential or temporal structure enters the graph construction, so we make no sequential-data claim.

A two-factor small-world screen. To make "destroyed" precise rather than a label, we screen every substrate on two graph statistics against degree-matched baselines—average clustering C (versus an Erdős–Rényi C_{rand} and a ring-lattice C_{latt}) and characteristic path length L (versus L_{rand})—combined into the Telesford index $\omega = L_{\text{rand}}/L - C/C_{\text{latt}}$ ($\omega < 0$ lattice-like, $\omega \approx 0$ small-world, $\omega \to 1$ random). Clean low-d manifolds sit firmly on the lattice side; the tangles do not:

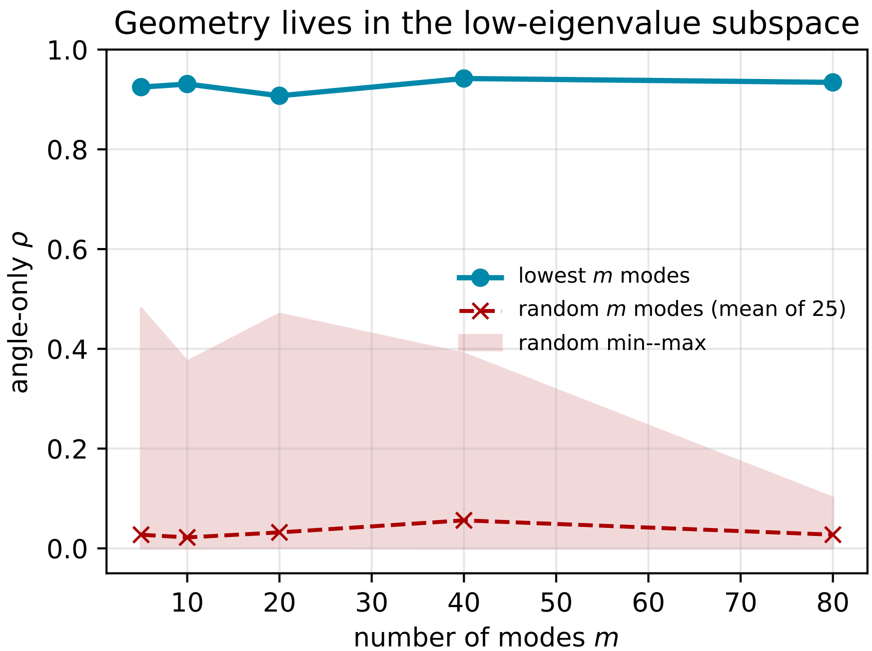

Figure 2: Geometry lives in the low-eigenvalue subspace: lowest-m modes preserve geodesics; a random-mode basis of equal size does not. The random curve is the mean over 25 draws and the shaded band its full min–max envelope—even the best-case random draw stays far below the low-mode curve.

substrate	angle-ρ	ω	dim-spread	verdict
2-torus	0.91	-0.55	0.14	clean
king-lattice	0.83	-0.50	0.51	clean
swiss-roll	0.93	-0.66	0.45	clean
king-lat. rewired	0.42	$+0.23$	0.77	small-world
causal set	0.57	$+0.75$	7.87	non-manifold

The screen robustly separates the genuine manifolds ($\omega \leq -0.50$) from strongly destroyed geometry ($\omega \geq +0.23$), and the inter-estimator dimension spread corroborates (≤ 0.51 clean versus ≥ 0.77 contaminated): the low angle-ρ on the rewired and Lorentzian substrates is small-world/non-manifold contamination, not a failure of the basis on genuine geometry. A king-lattice rewiring sweep both sharpens and honestly bounds this. As the rewire probability rises, angle-ρ collapses almost at once ($0.81 \to 0.47$ by 1% rewiring) while ω does not cross zero until $\approx 5\%$; so $\omega < 0$ is a *conservative* flag for destruction, and in the transition band (1–3%) angular fidelity is in fact the *more* sensitive detector of geometric corruption than either clustering or dimension spread. We therefore read the screen as confirming gross contamination, with angle-ρ itself the finer instrument—the mechanism behind the scaling-law outlier below.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1$–3.6),

$$\text{angle-}\rho \;\approx\; 1.09 - 0.157\,d, \qquad \text{slope } -0.157 \pm 0.028, \quad R^2 = 0.64, \quad d \in [1.0, 3.6].$$

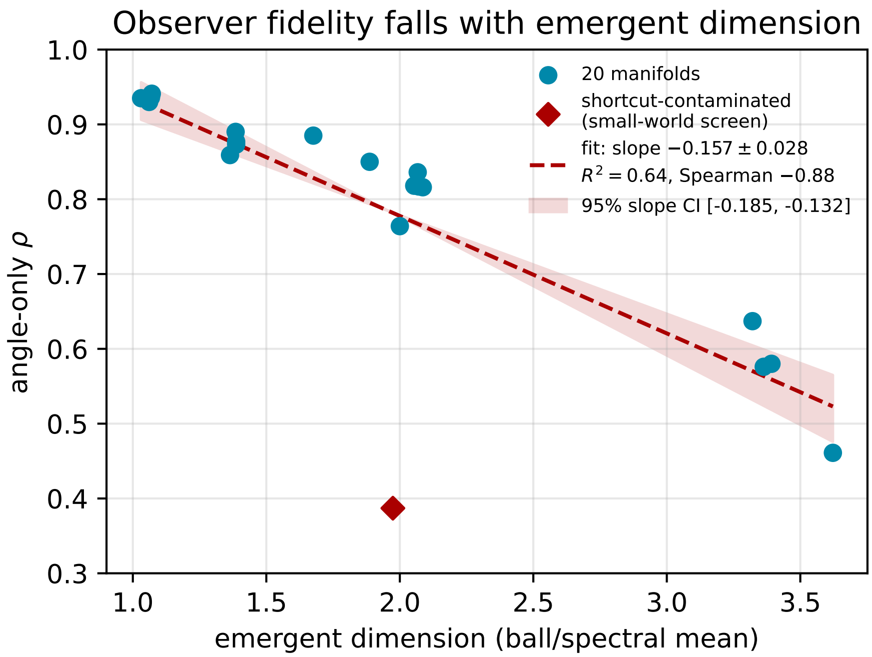

Figure 3: Angular fidelity falls linearly with emergent dimension across the 20 library manifolds (five Wolfram-model rules × four seeds, $d \approx 1$–3.6); the shaded band is the 95% bootstrap CI on the slope ($[-0.185, -0.132]$), drawn as its envelope about the data centroid. Per-point pair-resampling CIs are smaller than the plot markers ($\leq \pm 0.005$). The red diamond ($d{\approx}1.97$, $\rho{=}0.39$) is the single shortcut-contaminated manifold flagged by the small-world screen (§5.3), which drives the modest R^2.

(Spearman -0.881 [95% CI $-0.97, -0.67$; $n = 20$ manifolds] between ρ and d is a distribution-free monotonicity check; the linear fit's slope has a bootstrap 95% CI of $[-0.185, -0.132]$, comfortably excluding zero. The modest $R^2 = 0.64$ reflects a single outlier—a $d{\approx}1.97$ manifold at $\rho{=}0.39$—which the small-world screen of §5.3 flags as shortcut-contaminated rather than clean geometry.) The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \to 0$ fidelity, and the slope may vary with substrate roughness. A bounded model fits essentially as well and respects $0 < \rho < 1$: the logistic $\rho \approx \left[1 + e^{0.82\,(d-3.66)}\right]^{-1}$ gives $\rho(0) = 0.95$ and $\rho \to 0$ at $R^2 = 0.62$, against the linear 0.64. Exponential forms do not help—an unoffset Ae^{-kd} still has $A = 1.18 > 1$ and an offset version drives the asymptote negative. Within range the two are visually indistinguishable; we keep the linear coefficient as the local slope and note the logistic as the boundary-respecting global form. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

5.5 Verifying Corollary 1 and Conjecture 1

angle-ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.92$–$0.99$; radius-only $\rho \leq 0.08$). Corollary 1 holds cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \to 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an m-sweep at $n = 4000$, angle-ρ is flat ≈ 0.93 across $m = 5$–80 while the full embedding decays $0.773 \to 0.437$.

5.6 Which weighting carries the angle?

The direction depends on the per-mode weight, so we ablate it on the torus (d=2) and a 3-torus (d=3). Plain eigenmaps (w_k=1) give an angle that *collapses* with mode count—angle-ρ falls from 0.55 at m=10 to 0.06 at m=20 (d=2)—because equal weighting lets high modes swamp the direction. The commute weight $1/\sqrt{\lambda_k}$ is flat instead ($0.91 \to 0.89$), as is diffusion $e^{-\lambda_k t}$; both suppress high modes, which is what the geometry needs. The "flat in m" property is thus a property of the weighting, not of angles in general—and it selects the commute (or diffusion) weight, not the plain eigenmap. Degree correction, by contrast, is *invisible* to the angle: dividing or multiplying each row by $\sqrt{k_i}$ (the diffusion-map convention and its opposite) leaves angle-ρ unchanged to three digits (0.889 either way), because a per-node radial factor cancels under row-normalization—a direct confirmation that the degree lives in the radius (Corollary 1), not the angle.

5.7 A bake-off against named spectral distances

The ablation isolates the *weight*; a referee rightly asks how the angle fares against the standard *distances* built from the same spectrum. We score each on the identical graph and anchor set (Spearman ρ vs. unweighted geodesics), on a 2-torus (n=2000) and a 3-torus (n=3000).

distance	2-torus	3-torus
angle (ours)	**0.888**	**0.830**
full commute / resistance	0.601	0.360
biharmonic [27]	0.735	0.515
diffusion distance (best t) [5]	0.771	0.631
amplified commute [26]	0.628	0.364
Isomap, given the geodesics	*0.971*	*0.952*

The angle beats every purely spectral competitor, and the margin widens in d=3 where commute degeneracy bites harder. The most informative row is the amplified commute distance: von Luxburg, Radl and Hein designed it precisely to undo commute degeneracy by subtracting the $1/k_i+1/k_j$ term, yet it barely improves on raw commute (0.628 vs. 0.601) and stays far below the angle. *Subtracting* the degenerate radial term is not the same as *dividing it out*: row-normalization removes the whole degree-dependent radial factor at once, which is what the geometry needs (Lemma 1, Corollary 1). Isomap wins, but only because it is handed the geodesics as input—it is an oracle ceiling, not a competitor; the angle recovers 0.83–0.89 of that ceiling from the graph alone.

5.8 Radius is density, angle is geometry: a non-uniform torus

The decomposition "degree/density in the radius, geometry in the angle" predicts a sharp behaviour under *non-uniform* sampling that uniform substrates cannot test. We sample the flat 2-torus with density $p(x) \propto 1 + a\cos 2\pi x$, build a fixed-radius geometric graph, and sweep the contrast a (degree ratio up to 45× at a=0.9), scoring the angle against the *true* flat-torus geodesics—the intrinsic metric, which is unchanged by how we sample it. As predicted, the radius tracks the imposed density profile ($\rho(\|\Psi\|, 1/\sqrt{p}) = 0.53$–$0.67$), confirming it is a density coordinate. The raw angle, however, is *not* sampling-invariant: angle-ρ to the true metric degrades from 0.93 to 0.79 as the contrast grows, because the graph Laplacian converges to a density-weighted operator, not the Laplace–Beltrami operator [5, 28]. The classical fix recovers it exactly: the Coifman–Lafon α=1 normalization $W \mapsto D^{-1}WD^{-1}$ (with D the density estimate) removes the density weight, and the density-normalized angle is restored to 0.92–0.96, now *flat* across the whole contrast range. So the angle carries the geometry, but only once the density is properly quotiented out—and the same normalization that fixes the angle is the one that isolates the density into the radius.

5.9 Stability across substrate evolution

Universality across substrate *types* (§5) is a statement about fixed graphs. The angular basis also persists as a substrate *evolves*. For each of 151 qualifying Wolfram-model rules (95 arity-2, 56 arity-3) we embed successive rewrite snapshots on a fixed anchor set and compare, between consecutive snapshots, the Spearman persistence of the *angular* pairwise structure (`consec`) against the persistence of the underlying *geodesic* structure itself (`churn`)—the substrate's own rate of change is the baseline the basis must keep pace with. On all 143 rules resolvable at protocol scale (8 never yield ≥ 3 distinct snapshots and are excluded from the denominator but reported), the angular structure tracks the evolving geometry at least as stably as the graph's own geodesics, $\overline{\texttt{consec}} \geq 0.9\,\overline{\texttt{churn}}$: a $143/143 = 100\%$ pass against a pre-registered 80% bar, with zero substrate-unstable rules. Over the longest time-lever (first vs. last snapshot) angular persistence declines only gently with emergent dimension,

$$\text{slope } -0.029/\text{dim}, \quad \text{Spearman } -0.48, \quad d \in [0.98, 2.23],$$

and this replicates independently in both arities (arity-2 -0.028/dim, $\rho = -0.45$, $n = 95$; arity-3 -0.029/dim, $\rho = -0.54$, $n = 48$)—the same mechanism, unchanged by moving from binary to ternary rewriting.[2]

5.10 Continuum survival, and a dissociation

Every result so far is at finite N ($\approx$ 2–4k nodes, capped by dense `eigh`); the "uniform in N" clause of Corollary 1 and Conjecture 1 is untested at scale (§5.5). We test it directly by refining each substrate toward the continuum with a sparse eigensolver, $N \in \{2, 5, 10, 20, 50\} \times 10^3$, four seeds, $m = 10$ modes, $n = 300$ anchors—well past the $\sim$ 5k dense ceiling. The two substrate families dissociate. On genuine Riemannian substrates the basis *sharpens*: on random geometric graphs the angular fidelity is flat-to-rising in N, most cleanly in the higher-dimensional case,

$$\begin{aligned} \text{RGG } d{=}3: \quad & \text{angle-}\rho: \ 0.856 \to 0.912 \ \ (N: \ 2\text{k} \to 50\text{k}), \\ & \text{Spearman}(\text{angle-}\rho, \log N) = +1.00, \end{aligned}$$

with $d{=}2$ likewise monotone (+0.90, 0.916→0.929): refining the mesh makes the low modes resolve the geometry *better*, not worse, and both controls stay pinned (higher-mode $\rho \leq 0.05$, magnitude $\rho \leq 0.03$, eigensolver residual $\sim 10^{-14}$) at every N. The finite-N evidence of §§5–5.5 was not a small-graph artifact.

The dissociation is with *emergent* substrates. The $\mathrm{R}_{3\mathrm{D}}$ Wolfram-model manifold—geometry with no latent coordinates—*decays* monotonically with refinement, angle-$\rho: 0.749 \to 0.654$ (Spearman $= -1.00$), falling below a pre-registered floor of 0.75; and at the largest N its magnitude control begins to break the clean angle/radius separation that holds exactly on the RGGs (one 50k seed crosses $\rho_{\text{mag}} = 0.33$ and is gated out). A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not—its fine structure drifts away from a manifold as it grows. We read this not as a limitation but as a result: the angular basis is a *diagnostic* for whether a

[2] Pre-registered as Amendment 2: an earlier capture hook could append the same state once per crossed edge-count threshold, inflating consecutive stability toward a spurious 1.0. The corrected rule takes at most one snapshot per generation with dynamic 2× spacing; the numbers here are from that harness.

substrate's fine structure stays manifold-like under refinement, and it returns different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry.

One qualification is essential, and it deepens rather than weakens the reading: the dissociation is stated at a *fixed* observer budget ($m = 10$ while N grows $5\times$). Rerunning with m *spectral-gap-matched*—holding the resolved eigenvalue scale τ fixed, so m grows with N (to $m \approx 47$ at $N{=}10^4$)—inverts it: the RGG now *falls* ($0.856 \to 0.775$, Spearman $= -1.0$), because the added high modes inject sub-geodesic detail, while the emergent substrate stops decaying ($0.747 \to 0.692$, monotonicity gone, Spearman $= -0.5$). No single budget is optimal for both. So the dissociation is a property of the *fixed-budget* observer, not an absolute substrate invariant: a manifold keeps its geodesic geometry in a fixed low-mode band (refining sharpens those modes), whereas the emergent substrate spreads geometry across scales (a fixed observer resolves an ever-smaller fraction as it grows). "Clean geometry" is thus a relation between observer resolution and substrate scale—the sharpest form of the mode-budget consistency the angular basis exhibits (§5.5), and the motivation for a dimension-adaptive basis (§7). Modest evidence—one emergent substrate, one seed, $N \leq 10^4$—so we state the direction, not a law.[3]

5.11 A cross-domain probe: the moral embedding manifold

As a probe of whether the diagnostic transfers beyond synthetic and emergent graphs, we point it at a *semantic* space: 2,400 moral/ethical text items (BGE-M3, 1024-d), stratified across six traditions, in a $k{=}10$ nearest-neighbour graph. The metric here is the kNN graph's own hop geodesics, so this measures spectral reconstructability of that graph, not external semantic validity. All-pairs angle-$\rho = 0.90$ separates it sharply from destroyed controls (per-feature shuffle 0.23, isotropic Gaussian 0.31; higher-mode $\rho = 0.01$). Two confounds keep this honest, and both are real. First, the all-pairs number is partly cluster separation—the effect row-normalization is known to enhance [12]: decomposed, *between*-tradition pairs read 0.85 but *within*-tradition pairs, the ones that speak to local manifold structure, read only 0.67 (still above the controls, well below the all-pairs figure). Second, the informative magnitude is mostly hubness: $\rho(r_i, \text{kNN in-degree}) = 0.58$, so the radius tracks node degree (Corollary 1), not a moral-specific scale. We read the moral embedding as manifold-like *within* traditions and the cross-domain transfer as a graded diagnostic, not a headline; a generic-text control at matched n and model, k-sensitivity, and dataset documentation are deferred (§7).

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial-$\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

[3] Pre-registered (PREREG_RUNG4) before the run; survival bar angle-$\rho(N_{\max}) \geq \max(0.75,\, 0.9\,\text{angle-}\rho(N_{\min}))$ on $\geq 2/3$ substrates. Amendment 1, registered before any run: the whole-spectrum random-mode control needs a dense decomposition unavailable at $N \geq 50\text{k}$, so it is the top m of the lowest-64 sparse-computed modes—a faithful, sparse-affordable stand-in, same ≤ 0.15 bar.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [12], unifies it with the scale-invariant direction of KV-cache compression [10], and holds *universally* across substrates with genuine low-d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and sequential-trajectory substrates is evidence that scale-invariant angular coarse-graining is a general mechanism by which relational data carries geometric structure.

One interpretation is observer-theoretic. In Wolfram's observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making "the observer's basis" a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy- responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1 (in its rank form, and ideally the bi-Lipschitz strengthening); extending Corollary 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above $d \approx 2$; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

Reproducibility. The pre-registration documents (`PREREG_RUNG3/4` with dated amendments), all experiment scripts and per-run result files (the torus and universality benchmarks, the weighting ablation, the small-world screen, the continuum and scaled-budget refinements, the truncation/normalization checks, and the cross-domain probe), the manifold-rule sets behind the 20-point scaling law and the 151 evolution rules, and the random seeds and solver settings are released in the accompanying repository.

References

[1] M. Belkin and P. Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003.

[2] M. Belkin and P. Niyogi. Convergence of Laplacian eigenmaps. In *Advances in Neural Information Processing Systems 19*, 2007.

[3] N. García Trillos and D. Slepčev. A variational approach to the consistency of spectral clustering. *Applied and Computational Harmonic Analysis*, 45(2):239–281, 2018.

[4] P. W. Jones, M. Maggioni, and R. Schul. Universal local parametrizations via heat kernels and eigenfunctions of the Laplacian. *Annales Academiæ Scientiarum Fennicæ Mathematica*, 35:131–174, 2010.

[5] R. Coifman and S. Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1):5–30, 2006.

[6] J. Gorard. Some relativistic and gravitational properties of the Wolfram model. *Complex Systems*, 29(2):107–192, 2020.

[7] M. Gromov. Hyperbolic groups. In *Essays in Group Theory*, MSRI Publications 8, pp. 75–263, Springer, 1987.

[8] A. Ng, M. Jordan, and Y. Weiss. On spectral clustering: analysis and an algorithm. *Advances in Neural Information Processing Systems 14 (NeurIPS)*, 2002.

[9] M. Nickel and D. Kiela. Poincaré embeddings for learning hierarchical representations. *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

[10] I. Han, P. Kacham, A. Karbasi, V. Mirrokni, and A. Zandieh. PolarQuant: Quantizing KV caches with polar transformation. *arXiv preprint arXiv:2502.02617*, 2025. https://arxiv.org/abs/2502.02617. Accessed July 2026.

[11] R. Sarkar. Low-distortion delta-hyperbolic embedding in the hyperbolic plane. *Graph Drawing*, LNCS 7034, pp. 355–366, Springer, 2011.

[12] G. Schiebinger, M. J. Wainwright, and B. Yu. The geometry of kernelized spectral clustering. *Annals of Statistics*, 43(2):819–846, 2015.

[13] U. von Luxburg, M. Belkin, and O. Bousquet. Consistency of spectral clustering. *Annals of Statistics*, 36(2):555–586, 2008.

[14] U. von Luxburg, A. Radl, and M. Hein. Hitting and commute times in large random neighborhood graphs. *JMLR*, 15:1751–1798, 2014.

[15] S. Wolfram. *A New Kind of Science*. Wolfram Media, 2002.

[16] S. Wolfram. Finally we may have a path to the fundamental theory of physics…and it's beautiful. Writings blog, April 2020. https://writings.stephenwolfram.com/2020/04/finally-we-may-have-a-path-to-the-fundamental-theory-of-physics-and-its-beautiful/. Accessed July 2026.

[17] S. Wolfram. The concept of the ruliad. Stephen Wolfram Writings blog, November 2021. https://writings.stephenwolfram.com/2021/11/the-concept-of-the-ruliad/. Accessed July 2026.

[18] P. Bérard, G. Besson, and S. Gallot. Embedding Riemannian manifolds by their heat kernel. *Geometric and Functional Analysis (GAFA)*, 4(4):373–398, 1994.

[19] J. Bates. The embedding dimension of Laplacian eigenfunction maps. *Applied and Computational Harmonic Analysis*, 37(3):516–530, 2014.

[20] J. W. Portegies. Embeddings of Riemannian manifolds with heat kernels and eigenfunctions. *Communications on Pure and Applied Mathematics*, 69(3):478–518, 2016.

[21] J. Calder and N. García Trillos. Improved spectral convergence rates for graph Laplacians on ε-graphs and k-NN graphs. *Applied and Computational Harmonic Analysis*, 60:123–175, 2022.

[22] R. M. Rustamov. Laplace–Beltrami eigenfunctions for deformation invariant shape representation. *Proceedings of the Fifth Eurographics Symposium on Geometry Processing (SGP)*, pp. 225–233, 2007.

[23] J. Jin. Fast community detection by SCORE. *The Annals of Statistics*, 43(1):57–89, 2015.

[24] T. Qin and K. Rohe. Regularized spectral clustering under the degree-corrected stochastic blockmodel. *Advances in Neural Information Processing Systems 26 (NIPS)*, 2013.

[25] X. Cheng and G. Mishne. Spectral embedding norm: looking deep into the spectrum of the graph Laplacian. *SIAM Journal on Imaging Sciences*, 13(2):1015–1048, 2020.

[26] U. von Luxburg, A. Radl, and M. Hein. Getting lost in space: large sample analysis of the resistance distance. *Advances in Neural Information Processing Systems 23 (NIPS)*, 2010.

[27] Y. Lipman, R. M. Rustamov, and T. A. Funkhouser. Biharmonic distance. *ACM Transactions on Graphics*, 29(3):27, 2010.

[28] M. Hein, J.-Y. Audibert, and U. von Luxburg. Graph Laplacians and their convergence on random neighborhood graphs. *Journal of Machine Learning Research*, 8:1325–1368, 2007.